NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: ONE

CLASS: SS3 SUBJECT: Geography

TOPIC: World Regions: Africa

Introduction

- Regional geography studies the physical and human features of a region. Africa, the second largest continent (about 30 million km²), lies mostly in the tropics, crossed by the equator and bounded by the Atlantic and Indian Oceans, the Mediterranean and the Red Sea.

Physical features

- Three main landform zones: the plateaus and basins of the interior (Congo Basin, East African Plateau), the high mountain ranges (Atlas in the north, the Rift Valley highlands with Kilimanjaro, Kenya, Ruwenzori), and the low coastal plains. Great rivers: Nile (longest), Congo, Niger, Zambezi. Lakes: Victoria, Tanganyika, Malawi. Hot deserts: Sahara (north), Kalahari and Namib (south).

Climate and vegetation

- The equator gives high temperatures all year; rainfall controls vegetation belts: equatorial rainforest (Congo basin, West African coast), savanna (vast grasslands with scattered trees), desert (Sahara) and Mediterranean scrub (north and south tips), with montane vegetation in the highlands. Rainforests are disappearing through logging and farming (deforestation).

People and economic activities

- Africa's population is over 1.4 billion, with a youthful profile, rapid urban growth and high birth rates. Economic activities: subsistence and commercial agriculture (cocoa, coffee, cotton, oil palm), mining (gold, diamonds, oil), forestry, fishing and growing services. Problems: poverty, debt, food insecurity, poor infrastructure, conflict and the impacts of climate change (drought in the Sahel).

CLASS EVALUATION

1. Describe the three landform zones of Africa.
2. Name two rivers and lakes of Africa.
3. Explain how rainfall controls African vegetation.
4. State three economic problems facing African countries.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: TWO

CLASS: SS3 SUBJECT: Geography

TOPIC: Regional Geography: West Africa

Introduction

- West Africa is the region between the Sahara and the Gulf of Guinea (about 7.5 million km², 16 countries). Nigeria is the most populous, followed by Ghana, Ivory Coast, Senegal and the Niger Republic.

Physical and climatic features

- Landforms: the Fouta Djallon highlands (headwaters of many rivers), the Jos Plateau, the Adamawa highlands, coastal lowlands with lagoons and the Niger Delta. Rivers: Niger and Benue (navigable), Volta, Senegal, Gambia. Climate: the tropical continental (hot, dry north), tropical monsoon (south with heavy rains) and the Guinea-type coast; the alternation of the dry Harmattan winds and moist south-west monsoon controls the seasons and rainfall.

Vegetation

- Belts run roughly parallel to the coast: salt and freshwater swamp, tropical rainforest (with valuable timber: mahogany, obeche, iroko), Guinea savanna, Sudan savanna and Sahel (semi-desert). Overgrazing, logging and bush burning push the desert southward (desertification).

Economic life

- Agriculture employs the majority of people: cash crops (cocoa in Ghana and Ivory Coast; palm oil in Nigeria; groundnut in Senegal; cotton), food crops (yam, cassava, maize, rice), fishing along the coast and inland waters, mining (gold in Ghana, iron ore in Guinea, tin in Nigeria, oil in Nigeria), trading and industries around coastal ports (Lagos, Tema, Abidjan, Dakar). ECOWAS promotes regional integration.

CLASS EVALUATION

1. Name four West African countries.
2. Which winds control the seasons of West Africa?
3. Describe the vegetation belts from coast to Sahel.
4. Why is cocoa important to Ghana and Ivory Coast?

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: THREE

CLASS: SS3 SUBJECT: Geography

TOPIC: Comparative Studies (Nigeria & Ghana)

Introduction

- Comparative geography examines the similarities and differences between two or more regions — usually in position, relief, climate, vegetation, population and economic development. This week compares Nigeria with Ghana (both British colonies, both oil/gold producers, both cocoa exporters).

Comparison

- **Size/population:** Nigeria 923,000 km², 200m+ people; Ghana 238,500 km², about 33m — Nigeria is four times larger and far more populous. **Relief:** both have coastal plains, plateaus and river basins (Niger–Benue; Volta basin). **Climate:** both tropical, with the south wetter than the north; Ghana's coast has the drier 'dry equatorial' zone due to the cold Canary current. **Vegetation:** both run rainforest → savanna → Sahel. **Resources:** Nigeria — oil dominant; Ghana — gold, cocoa, and now oil and bauxite. **Economy:** both depend on primary products, with growing services.

Conclusions of comparison

- Comparisons show how physical environment and history shape development; they help planners learn from neighbours and expose the sources of regional imbalance. Nigeria's oil wealth has not yet produced the per-capita income of smaller, better-managed neighbours.

CLASS EVALUATION

1. Compare the size and population of Nigeria and Ghana.
2. Explain one similarity and one difference in their climate.
3. Compare their main exports.
4. What lessons can Nigeria learn from Ghana's economy?

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FOUR

CLASS: SS3 SUBJECT: Geography

TOPIC: Map Interpretation

Introduction

- Topographical maps represent the earth's surface — relief, drainage, vegetation, settlement and human activities — using conventional signs, contours and a scale. Interpreting them is a core exam skill.

Key elements

- **Scale** — the ratio between map and ground distance (1:50,000 means 1 cm = 500 m). Use the linear scale or the scale statement; distances are measured in straight lines (with a ruler) or along roads (with a string/map measurer).
- **Contours** — lines joining points of equal height above sea level: close contours = steep slope, wide spacing = gentle slope, concentric circles = hills/peaks, V-shapes pointing uphill = valleys, pointing downhill = spurs.
- **Conventional signs** — colours (blue water, green vegetation, brown relief, red roads) and symbols (bridges, churches, airports, wells, settlements); grid lines and grid references (4- and 6-figure) fix positions.

Interpretation skills

- Describe relief (high/low land, slopes), drainage (rivers, direction of flow), settlement patterns (dispersed, nucleated, linear), land use (farming, mining, forestry, towns), and the relationship between the physical environment and human activity. Calculated answers: gradient = vertical interval ÷ horizontal distance; density of settlement; bearing between points.

CLASS EVALUATION

1. What is the distance on the ground of 6 cm on a 1:50,000 map?
2. How do contours show a steep slope?
3. What do V-shaped contour patterns pointing uphill indicate?
4. Describe the physical and human features of a map area from a given extract.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FIVE

CLASS: SS3 SUBJECT: Geography

TOPIC: Weather & Climate (Revision & Application)

Introduction

- Weather is the state of the atmosphere at a place at a particular time; climate is the average weather of a place over many years (30+). This week consolidates the instruments and the concepts behind weather maps.

Elements and instruments

- **Temperature** — thermometer (maximum–minimum, six's), in a Stevenson screen.
- **Rainfall** — rain gauge (funnel + measuring cylinder), recorded daily; the total over a month or year is the rainfall total.
- **Pressure** — barometer (mercury, aneroid); isobars join equal pressures on maps.
- **Wind** — anemometer (speed), wind vane (direction); **Humidity** — hygrometer/wet-and-dry bulb; **Sunshine** — Campbell–Stokes recorder; **Clouds** — observation; **Evaporation** — evaporimeter.

Reading weather data

- Mean daily temperature = (max + min) ÷ 2; diurnal range = max – min; annual range for a station from the monthly means. Describe climate from a climatic graph: hot and wet all year (equatorial); hot with summer rain (tropical wet and dry); cool winters (Mediterranean). The Nigerian climate graph shows two seasons — wet (April–October) and dry (November–March).

Application

- Climate shapes agriculture (planting seasons), settlement, transport, health (malaria in wet zones) and economic planning. Nigerians plan planting around the onset of rains; the Harmattan brings dry, dusty days in the north.

CLASS EVALUATION

1. Define weather and climate.
2. Name the instrument for rainfall and how it is read.
3. Calculate the mean daily temperature: max 32 °C, min 21 °C.

4. Describe the two seasons of Nigeria.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL, RUMOMASI

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: SIX

CLASS: SS3 SUBJECT: Geography

TOPIC: Environmental Hazards & Conservation

Introduction

- Environmental hazards are natural or man-made events that threaten life and property. In Nigeria the major hazards are flooding, desertification, erosion, drought, deforestation and pollution — each worsened by human activity.

Major hazards in Nigeria

- **Flooding** — heavy rains and blocked drains (Lagos, Benin, Lokoja, 2012 and 2022 floods); rivers overflow; slums on flood plains suffer most.
- **Desertification** — the Sahara advances south because of overgrazing, bush burning and tree felling in the Sahel (hits Borno, Yobe, Sokoto, Katsina).
- **Soil erosion** — gully erosion in the east (Anambra, Imo) and sheet erosion elsewhere, from deforestation and poor farming methods.
- **Drought** — prolonged rain failure in the north, causing crop failure and famine. **Deforestation** — loss of forest cover for fuel and farmland. **Pollution** — oil spills in the Delta, industrial waste, refuse and air pollution.

Causes and effects

- Causes are mainly human: farming on marginal land, logging, overgrazing, poor urban planning, dumping of waste, gas flaring, construction on flood plains. Effects: loss of lives, homes and farmland; food shortage; siltation of rivers; loss of biodiversity; displacement of people; economic losses.

Conservation measures

- Afforestation and reforestation (the Great Green Wall across the Sahel); contour ploughing, terracing and cover cropping; controlled grazing; construction of drainage and flood control (dredging of rivers); tree planting campaigns; waste management (recycling); environmental laws and agencies (NESREA, forestry departments); public education and environmental impact assessment.

CLASS EVALUATION

1. Define environmental hazard.

2. State three causes of desertification in the Sahel.
3. List two effects of gully erosion.
4. Mention three measures for conserving Nigeria's environment.